

JEE MAIN REVISION MASTER: ATOMS AND NUCLEI SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 284 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Nuclear Physics, Binding Energy & Radioactivity	149 PYQs	Nuclear Physics
Very High	Bohr's Hydrogen Model & Spectral Series	135 PYQs	Atomic Models

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Nuclear Physics, Binding Energy & Radioactivity	$R = R_0 A^{1/3}$, $\Delta E_b = [Z m_p + (A - Z)m_n - M_{\text{nuc}}]c^2$, $N(t) = N_0 e^{-\lambda t}$, $T_{1/2} = \frac{0.693}{\lambda} \approx 0.693\tau$
Bohr's Hydrogen Model & Spectral Series	$r_n = 0.529 \frac{n^2}{Z} \text{ \AA}$, $v_n = 2.18 \times 10^6 \frac{Z}{n} \text{ m/s}$, $E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$, $\frac{1}{\lambda} = R Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Nuclear Physics, Binding Energy & Radioactivity (Very High Yield — 149 PYQs)

Core Principle: Nuclear radius scaling $R \propto A^{1/3}$, constant nuclear density, mass defect conversion ($1 \text{ amu} \approx 931.5 \text{ MeV}$), and radioactive decay law.

■ **Caution / JEE Pitfall:** Nuclear density is CONSTANT ($\approx 2.3 \times 10^{17} \text{ kg/m}^3$) and completely independent of the mass number A !

■ **Speed Trick:** Amount remaining after n half-lives: $N = (N_0 / 2^n)$, where $n = t / T_{1/2}$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (Numerical): "The average energy released per fission for the nucleus of ${}_{92}^{235}\text{U}$ is 190 MeV. When all the atoms of 47 g pure ${}_{92}^{235}\text{U}$ undergo..." [Ans: 228]
- [2026 Evening Shift] Q#2 (MCQ): "Assuming the experimental mass of ${}_{6}^{12}\text{C}$ as $12u$, the mass defect of ${}_{6}^{12}\text{C}$ atom is ____ MeV/c^2 . (Mass of proton $= 1.0...$ " [Ans: B]
- [2026 Evening Shift] Q#3 (MCQ): "Two radioactive substances A and B of mass numbers 200 and 212 respectively, shows spontaneous α -decay with same Q value of 1 MeV ..." [Ans: C]

3.2 Bohr's Hydrogen Model & Spectral Series (Very High Yield — 135 PYQs)

Core Principle: Quantization of angular momentum $L = n\hbar$, orbital radii, energy levels of hydrogenic species (Z), and emission lines.

■ **Caution / JEE Pitfall:** Balmer series lines fall in the VISIBLE spectrum ($n_1 = 2$). Lyman series lines fall in the ULTRAVIOLET ($n_1 = 1$).

■ **Speed Trick:** Total number of spectral lines emitted when electron de-excites from state n to ground state is $N = (n(n-1) / 2)$.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "The ratio of momentum of the photons of the 1st and 2nd line of Balmer series of Hydrogen atoms is α/β " [Ans: D]
- [2026 Evening Shift] Q#1 (MCQ): "Which of the following pair of nuclei are isobars of the element?..." [Ans: A]
- [2026 Morning Shift] Q#2 (MCQ): "In hydrogen atom spectrum, ($R \rightarrow$ Rydberg's constant) A. the maximum wavelength of the radiation of Lyman series is $\frac{4}{3R}$..." [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

The ratio of momentum of the photons of the 1st and 2nd line of Balmer series of Hydrogen atoms is α/β . The possible values of α and β are:-

- (A) 27 and 20
- (B) 3 and 16
- (C) 5 and 36
- (D) 20 and 27 [Correct]

Correct Answer: D

Solution Analysis: According to the de Broglie relation, the momentum (p) of a photon is inversely proportional to its wavelength (λ): $p = (h / \lambda)$ Therefore, the ratio of momenta is the inverse ratio of their wavelengths:

$\frac{p_1}{p_2} = \frac{\lambda_2}{\lambda_1}$ The Balmer series occurs when an electron transitions from a higher energy level (n_2) to the second energy level ($n_1 = 2$). The wavelength is given by Rydberg formula: $(1 / \lambda) = R(\frac{1}{2^2} - \frac{1}{n^2})$ For 1st line of Balmer transition from $n=3$ to $n=2$. $\frac{1}{\lambda_1} = R(\frac{1}{2^2} - \frac{1}{3^2})$

Benchmark Problem 2: JEE Main 2026 (Online) 23rd January Evening Shift (Q#1)

The average energy released per fission for the nucleus of ${}_{92}^{235}\text{U}$ is 190 MeV . When all the atoms of 47 g pure ${}_{92}^{235}\text{U}$ undergo fission process, the energy released is $\alpha \times 10^{23}\text{MeV}$. The value of α is ____ . (Avogadro Number = 6×10^{23} per mole)

Correct Answer: 228

Solution Analysis: The number of moles (n) is given by the formula : $n = (\text{Given Mass} / \text{Molar Mass})$ Given : Mass of Uranium = 47g Molar Mass of ${}_{92}^{235}\text{U} = 235\text{g/mol}$ $n = (47 / 235) = (1 / 5) = 0.2\text{moles}$ Each mole contains a number of atoms equal to Avogadro's Number (N_A) . $N = n \times N_A \Rightarrow N = 0.2 \times (6 \times 10^{23})$ Since each atom (nucleus) releases 190 MeV upon fission : $E = N \times \text{Energy per fission} \Rightarrow E = (1.2 \times 10^{23}) \times 190\text{MeV} \Rightarrow E = 228 \times 10^{23}\text{MeV} \Rightarrow \alpha = 228$